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22CH203 – ENGINEERING CHEMISTRY II

UNIT – III & METAL CORROSION AND ITS PREVENTION

Oxidation of metals: Electrochemical origin of corrosion – electromigration - electron transfer in the presence and absence of moisture - galvanic series. Strategies for corrosion control: Galvanic anode and impressed current.

LESSON PLAN 2:

Electron transfer in the presence of moisture - galvanic series.

TOPIC - ELECTROCHEMICAL CORROSION AND ITS MECHANISM

1. WET CORROSION OR ELECTROCHEMICAL CORROSION:

Wet corrosion occurs under the following conditions:

(i) When two dissimilar metals or alloys are in contact with each other in the presence of an aqueous solution or moisture.

(ii) When a metal is exposed to varying concentration of oxygen or any electrolyte.

1.1 MECHANISM OF WET CORROSION:

Under the above conditions, one part of the metal becomes anode and another part becomes cathode.

1. At anode: oxidation (or) dissolution of metal occurs

$$M \longrightarrow M^{2+} + 2e^-$$

2. At cathode: reduction reaction occurs, which depends on nature of the corrosive environment.

a) Acidic environment:

If the corrosive environment is *acidic, hydrogen* evolution occurs at cathodic part.

$$2H^+ + 2e^- \longrightarrow H_2\uparrow$$

b) Neutral/Alkaline environment:

If the corrosive environment is *slightly alkaline (or) neutral,* hydroxide ions are formed at cathodic part.

$$\tfrac{1}{2}O_2 + 2e^- + H_2O \longrightarrow 2OH^-$$

Thus, the metal ions (from anodic part) and non-metallic ions (from cathodic part) diffuse towards each other through conducting medium and form a corrosion product between anode and cathode.

1.1.1 Hydrogen Evolution Type Corrosion:

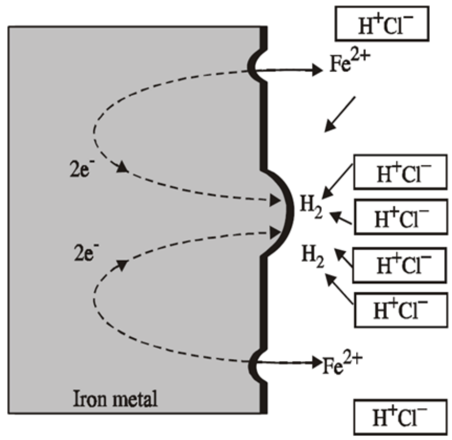

Figure 1.1 Hydrogen evolution type corrosion

All metals above hydrogen in the electrochemical series have a tendency to get dissolved in acidic solution with simultaneous evolution of hydrogen gas. When iron metal contacts with non-oxidizing acid like HCl, H_2 evolution occurs.

At Anode: Iron undergoes dissolution to give Fe^{2+} ions with the liberation of electrons.

$$Fe \longrightarrow Fe^{2+} + 2e^- \quad \text{(oxidation)}$$

At Cathode: The liberated electrons flow from anodic to cathodic part, where H^+ ions get reduced to H_2.

$$2H^+ + 2e^- \longrightarrow H_2\uparrow \quad \text{(reduction)}$$

1.1.2 Absorption of Oxygen Type Corrosion:

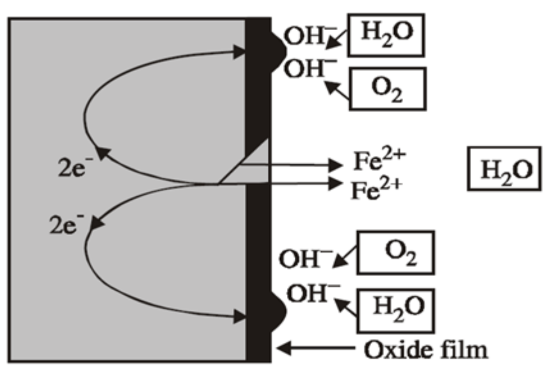

Figure 1.2 Absorption of oxygen type corrosion

The surface of iron is usually, coated with a thin film of iron oxide. However, if the oxide film grows, some crack will form and anodic areas are created on the surface while the remaining part acts as cathode.

When iron metal contacts a neutral or slightly alkaline solution of an electrolyte in presence of oxygen, OH^- ions are formed.

At Anode

Iron dissolves as Fe2+ with the liberation of electrons.

$$Fe \longrightarrow Fe^{2+} + 2e^- \quad \text{(oxidation)}$$

At Cathode:

$$\tfrac{1}{2}O_2 + H_2O + 2e^- \longrightarrow 2OH^-$$

Thus the net cell reaction is

$$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2\downarrow$$

If enough O_2 is present, $Fe(OH)_2$ is easily oxidized to $Fe(OH)_3$, a rust $Fe_2O_3.H_2O$.

$$4Fe(OH)_2 + O_2 + 2H_2O \longrightarrow 4Fe(OH)_3$$

The liberated electrons flow from anodic to cathodic part through metal, where the electrons are taken up by the dissolve oxygen to form OH^- ions.

2. DIFFERENCE BETWEEN CHEMICAL AND ELECTROCHEMICAL CORROSION:

S.No	Chemical corrosion	Electrochemical corrosion
1	It occurs only in dry corrosion	It occurs in the presence of moisture or electrolyte
2	It is due to the direct chemical attack on the metal by environment	It is due to the setup of a large number of anodic and cathodic areas
3	Even a homogenous metal surface gets corroded	Heterogeneous surface of bimetallic surface is the condition
4	Corrosion products accumulate in the same place where corrosion occurs	Corrosion occurs at the anode where corrosion products are formed elsewhere
5	Chemical corrosion is self-controlled	Electrochemical corrosion is a continuous process
6	It follows adsorption mechanism. *Example: Formation of mild scale on iron surface*	It follows electrochemical reaction. *Example: Rusting of iron in moist atmosphere.*

Discussion Questions:

1. Compare the corrosion of metals based on the presence and absence of moisture.
2. Predict the effect of nature of electrolyte on the mechanism of corrosion.